

Exercise JA

Q1. The value of $\cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right)$ is

- (A) $\frac{23}{25}$ (B) $\frac{25}{23}$ (C) $\frac{23}{24}$ (D) $\frac{24}{23}$

Ans. B

Sol.

$$\begin{aligned} & \cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \sum_{k=1}^n 2k\right)\right) \\ &= \cot\left(\sum_{n=1}^{23} \cot^{-1}\left(1 + \frac{2n(n+1)}{2}\right)\right) \\ &= \cot\left(\sum_{n=1}^{23} \tan^{-1} \frac{1}{1+n(n+1)}\right) \\ &= \cot\left(\sum_{n=1}^{23} \tan^{-1} \frac{(n+1) - n}{1+n(n+1)}\right) \\ &= \cot\left(\sum_{n=1}^{23} (\tan^{-1}(n+1) - \tan^{-1}n)\right) \\ &= \cot(\tan^{-1} 24 - \tan^{-1} 1) \\ &= \cot\left(\tan^{-1} \frac{23}{25}\right) = \cot^{-1}\left(\cot^{-1}\left(\frac{25}{23}\right)\right) = \frac{25}{23} \end{aligned}$$

Q2. Match List I with List II and select the correct answer using the code given below the lists :

List - I

List

- II

P. $\left(\frac{1}{y^2} \left(\frac{\cos(\tan^{-1} y \sin(\tan^{-1} y))}{\cot(\sin^{-1} y) + (\tan(\sin^{-1} y))}\right)^2 + y^4\right)^{1/2}$ takes value

1. $\frac{1}{2}\sqrt{\frac{5}{3}}$

Q. If $\cos x + \cos y + \cos z = 0 = \sin x + \sin y + \sin z$ then possible

2. $\sqrt{2}$

value of $\cos \frac{x-y}{2}$ is

R. If $\cos\left(\frac{\pi}{4} - x\right) \cos 2x + \sin x \sin 2x \sec x = \cos x \sin 2x \sec x$

3. $\frac{1}{2}$

+ $\cos\left(\frac{\pi}{4} + x\right) \cos 2x$ then possible value of $\sec x$ is

S. If $\cot(\sin^{-1} \sqrt{1-x^2})$ then $\sin(\tan^{-1}(x\sqrt{6})), x \neq 0$ possible value of x is

4. 1

- (A)

P	Q	R	S
4	3	1	2

 (B) 4 3 2 1 (C) 3 4 2 1 (D) 3 4 1 2

Ans. B

Sol.

$$(P) \left(\frac{1}{y^2} \left(\frac{\frac{1}{\sqrt{1+y^2}} + \frac{y^2}{\sqrt{1+y^2}}}{\frac{\sqrt{1-y^2}y}{y} + \frac{y}{\sqrt{1-y^2}}} \right)^2 + y^4 \right)^{1/2}$$

$$\Rightarrow \left(\left(\frac{(1+y^2)y^2(1-y^2)}{y^2} \right) + y^2 \right)^{1/2}$$

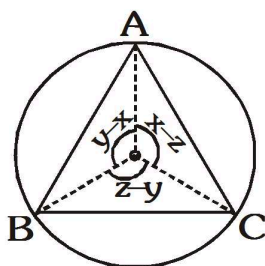
$$\Rightarrow \left((1-y^4) + y^4 \right)^{1/2}$$

$$\Rightarrow \left((1-y^4) + y^4 \right)^{1/4} = 1$$

(Q) A (cos x, sin x), B(cos y, sin y) & C (cos z, sin z) lie on circle $x^2 + y^2 = 1$

$\therefore (0, 0)$ is circumcentre as well as centroid of $\triangle ABC$

$\Rightarrow \triangle ABC$ is an equilateral triangle



$$y - x = \frac{2\pi}{3}$$

$$\cos \frac{y-x}{2} = \cos \frac{\pi}{3} = \frac{1}{2}$$

$$(R) \cos 2x \left(\cos \left(\frac{\pi}{4} - x \right) - \cos \left(\frac{\pi}{4} + x \right) \right)$$

$$= \sin 2x (-\tan x)$$

$$\sqrt{2} \sin x \cos 2x = \sin 2x (1 - \tan x)$$

$$\sin x \left(\sqrt{2} \cos 2x - 2(\sin x - \cos x) \right) = 0$$

$$\Rightarrow \sin x = 0 \text{ or } \sin x = \cos x \text{ or } \sin x + \cos x = \sqrt{2}$$

$$\Rightarrow \sec x = 1 \text{ or } \sec x = \pm \sqrt{2}$$

$$(S) \cot \left(\sin^{-1} \sqrt{1-x^2} \right) = \sin \left(\tan^{-1}(x\sqrt{6}) \right)$$

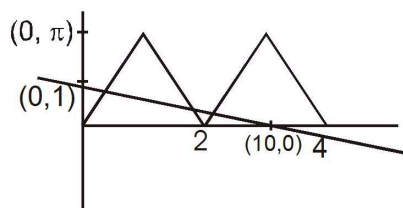
$$\Rightarrow 1 - 6x^2 = 6 + 6x^2$$

$$\Rightarrow x = \sqrt{\frac{5}{12}} = \frac{\sqrt{5}}{2\sqrt{3}}$$

Q3. Let $f : [0, 4\pi] \rightarrow [0, \pi]$ be defined by $f(x) = \cos^{-1}(\cos x)$. The number of points $x \in [0, 4\pi]$ satisfying the equation $f(x) = \frac{10-x}{10}$ is :

Ans. 3

Sol. $f(x) = \cos^{-1}(\cos x) \in x [0, 4\pi]$ & $f(x) = \frac{10-x}{10} = 1 - \frac{x}{10}$



so, 3 solution.

Q4. If $\alpha = 3 \sin^{-1} \left(\frac{6}{11} \right)$ and $\beta = 3 \cos^{-1} \left(\frac{4}{9} \right)$ where the inverse trigonometric functions take only the principal values, then the correct option (s) is (are) :

(A) $\cos \beta > 0$ (B) $\sin \beta < 0$ (C) $\cos (\alpha + \beta) > 0$ (D) $\cos \alpha < 0$

Ans. D, B, C

Sol. $\alpha = 3 \sin^{-1} \frac{6}{11} > 3 \sin^{-1} \frac{6}{12}$ and $\beta = 3 \cos^{-1} \frac{4}{9} > 3 \cos^{-1} \frac{4}{8}$

$$\Rightarrow \alpha > \frac{\pi}{2} \cdot 3 \text{ \& } \beta > \pi$$

$$\Rightarrow \alpha + \beta > \frac{3\pi}{2}$$

Q5. The number of real solutions of the equation

$$\sin^{-1} \left(\sum_{i=1}^{\infty} x^{i+1} - x \sum_{i=1}^{\infty} \left(\frac{x}{2} \right)^i \right) = \frac{\pi}{2} - \cos^{-1} \left(\sum_{i=1}^{\infty} \left(-\frac{x}{2} \right)^i - \sum_{i=1}^{\infty} (-x)^i \right)$$

lying in the interval $\left(-\frac{1}{2}, \frac{1}{2} \right)$ is _____. (Here, the inverse trigonometric functions $\sin^{-1}x$

and $\cos^{-1}x$ assume values in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$ and $[0, \pi]$, respectively.)

Ans. 2

Sol. $\lim_{n \rightarrow \infty} \sin^{-1} \left(\sum_{i=1}^n x^{i+1} - x \sum_{i=1}^n \left(\frac{x}{2} \right)^i \right) = \frac{\pi}{2} \lim_{n \rightarrow \infty} \cos^{-1} \left(\sum_{i=1}^n \left(-\frac{x}{2} \right)^i - \sum_{i=1}^n (-x)^i \right)$

$$\left(\frac{x^2}{1-x} - x \frac{\frac{x}{2}}{1-\frac{x}{2}} \right) = \frac{x}{1+x} + \frac{\left(-\frac{x}{2} \right)}{1+\frac{x}{2}}$$

$$\frac{x^2}{1-x} - \frac{x^2}{2-x} = \frac{x}{1+x} - \frac{x}{2+x}$$

$$\frac{x^2}{1-x} - \frac{x}{1+x} = \frac{x^2}{2-x} - \frac{x}{2+x}$$

$$\frac{x(1+x) - (1-x)}{1-x^2} = \frac{2x+x^2 - 2x-1}{4-x^2} \text{ or } x=0$$

$$\frac{x^2 + 2x - 1}{1-x^2} = \frac{x^2 + 3x - 2}{4-x^2}$$

$$\Rightarrow x^3 + 2x^2 + 5x - 2 = 0$$

$$\text{Let } f(x) = x^3 + 2x^2 + 5x - 2$$

$$f'(x) > 0$$

$$f(0) = -2 \text{ and } f(1/2) = 9/8 \text{ so one root in } \left(0, \frac{1}{2}\right)$$

$$\Rightarrow 2 \text{ roots}$$

Q6. For any positive integer n , define $f_n : (0, \infty) \rightarrow \mathbb{R}$ as

$$f_n(x) = \sum_{j=1}^n \tan^{-1} \left(\frac{1}{1 + (x+j)(x+j-1)} \right) \text{ for all } x \in (0, \infty).$$

(Here, the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.)

Then, which of the following statement(s) is (are) TRUE?

$$(A) \sum_{j=1}^5 \tan^2(f_j(0)) = 55 \quad (B) \sum_{j=1}^{10} (1 + f'_j(0)) \sec^2(f_j(0)) = 10$$

$$(C) \text{ For any fixed positive integer } n, \lim_{x \rightarrow \infty} \tan(f_n(x)) = \frac{1}{n}$$

$$(D) \text{ For any fixed positive integer } n, \lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$$

Ans. D

$$\text{Sol. } f_n(x) = \sum_{i=1}^{10} \tan^{-1} \left(\frac{1}{1 + (x+j)(x+j-1)} \right)$$

$$f_n(x) = \tan^{-1}(x+n) - \tan^{-1}(x) \Rightarrow f'_n(x) = \frac{1}{1+(x+n)^2} - \frac{1}{1+x^2}$$

$$f_n(0) = \tan^{-1}(n) \Rightarrow \tan^2(\tan^{-1} n) = n^2$$

$$(A) \sum_{j=1}^5 \tan^2(f_j(0)) = \sum_{j=1}^5 j^2 = \frac{5 \cdot 6 \cdot 11}{6} = 55$$

$$(B) f'_n(0) = \frac{1}{1+n^2} - 1 \Rightarrow 1 + f'_n(0) = \frac{1}{1+n^2}$$

$$\sec^2(f_n(0)) = \sec^2(\tan^{-1}(n)) = 1 + n^2.$$

$$\text{Hence } (1 + f_n(0)) = \left(\frac{1}{1 + n^2} \right) (1 + n^2) = 0$$

$$\text{So } \sum_{j=1}^{10} (1 + f'_i(0)) \sec^2(f_i(0)) = \sum_{i=1}^{10} 1 = 10$$

$$\lim_{x \rightarrow \infty} f_n(x) = \lim_{x \rightarrow \infty} \tan^{-1} \left(\frac{n}{1 + x(n + x)} \right) = 0$$

$$\lim_{x \rightarrow \infty} \tan(f_n(x)) = 0 \quad \& \quad \lim_{x \rightarrow \infty} \sec^2(f_n(x)) = 1$$

Q7. Let. $E_1 = \left\{ x \in \mathbb{R} : x \neq 1 \text{ and } \frac{x}{x-1} > 0 \right\}$ and

$$E_2 = \left\{ x \in E_1 : \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right) \text{ is a real number} \right\}.$$

(Here, the inverse trigonometric function $\sin^{-1}x$ assumes values in $\left[-\frac{\pi}{2}, \frac{\pi}{2} \right]$.)

Let $f : E \rightarrow \mathbb{R}$ be the function defined by $f(x) = \log_e \left(\frac{x}{x-1} \right)$ and $g : E_2 \rightarrow \mathbb{R}$ be the function defined by $g(x) = \sin^{-1} \left(\log_e \left(\frac{x}{x-1} \right) \right)$

LIST-I

P. The range of f is

Q. The range of g contains

R. The domain of f contains

S. The domain of g is

LIST-II

$$1. \left(-\infty, \frac{1}{1-e} \right] \cup \left[\frac{e}{e-1}, \infty \right)$$

$$2. (0, 1)$$

$$3. \left[-\frac{1}{2}, \frac{1}{2} \right]$$

$$4. (-\infty, 0) \cup (0, \infty)$$

$$5. \left(-\infty, \frac{e}{e-1} \right]$$

$$6. (-\infty, 0) \cup \left(\frac{1}{2}, \frac{e}{1-e} \right]$$

The correct option is:

$$(A) P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 1 \quad (B) P \rightarrow 3; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$$

$$(C) P \rightarrow 4; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 6 \quad (D) P \rightarrow 4; Q \rightarrow 3; R \rightarrow 6; S \rightarrow 5$$

Ans. A

$$\text{Sol. } E_1 : \frac{x}{x-1} > 0 \quad \Rightarrow \quad x \in (-\infty, 0) \cup (1, \infty)$$

$$E_2 : -1 \leq \ln \left(\frac{x}{x-1} \right) \leq 1 \quad \Rightarrow \quad \frac{1}{e} \leq \frac{x}{x-1} \leq e \quad \Rightarrow \quad \frac{1}{e} \leq 1$$

$$\frac{1}{e} - 1 \leq \frac{1}{x-1} \leq e - 1 \quad \Rightarrow \quad (x-1) \in \left(-\infty, \frac{e}{1-e} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

$$x \in \left(-\infty, \frac{e}{e-1} \right] \cup \left[\frac{1}{e-1}, \infty \right)$$

$$\text{Now } \frac{x}{x-1} \in (1, \infty) - \{1\} \quad \forall x \in E_1 \quad \Rightarrow \quad \ln \left(\frac{x}{x-1} \right) \in (-\infty, \infty) - \{0\}$$

$$\sin^{-1} \left(\left(\ln \frac{x}{x-1} \right) \right) \in \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] - \{0\}$$

Q8. For non negative integers n let

$$f(n) = \frac{\sum_{k=1}^n \sin \left(\frac{k+1}{n+2} \pi \right) \sin \left(\frac{k+2}{n+2} \pi \right)}{\sum_{k=0}^n \sin^2 \left(\frac{k+1}{n+2} \pi \right)}$$

Assuming $\cos^{-1} x$ takes values in $[0, \pi]$ which of the following options is/are correct

$$(A) f(4) = \frac{\sqrt{3}}{2} \quad (B) \text{ If } \alpha = \tan(\cos^{-1} f(6)) \text{ then } \alpha^2 + 2\alpha - 1 = 0 \quad (C) \lim_{n \rightarrow \infty} f(n) = \frac{1}{2}$$

$$(D) \sin(7\cos^{-1} f(5)) = 0$$

Ans. A, B, D

Sol.

$$f(n) = \frac{\sum_{k=0}^n 2 \sin \left((k+1) \frac{\pi}{n+2} \right) \sin \left((k+2) \frac{\pi}{n+2} \right)}{\sum_{k=0}^n 2 \sin^2 \left((k+1) \frac{\pi}{n+2} \right)}$$

$$\frac{\sum_{k=0}^n \left(\cos \frac{\pi}{n+2} - \cos \left((2k+3) \frac{\pi}{n+2} \right) \right)}{\sum_{k=0}^n -\cos \left((2k+2) \frac{\pi}{n+2} \right)}$$

$$f(n) = \frac{(n+1) \cos \left(\frac{\pi}{n+2} \right) - \sum_{k=0}^n \left((2k+3) \frac{\pi}{n+2} \right)}{(n+1) \sum_{k=0}^n \cos \left((2k+2) \frac{\pi}{n+2} \right)} \quad \dots\dots\dots(1)$$

$$\sum_{k=0}^n \cos \left((2k+3) \frac{\pi}{n+2} \right) = \cos \left(\frac{3\pi}{n+2} \right) + \cos \frac{5\pi}{n+2} + \cos \frac{5\pi}{n+2} + \cos \frac{7\pi}{n+2} + \dots\dots + (2n+3) \frac{\pi}{n+2} + \dots\dots +$$

$$(2n+3) \frac{\pi}{2n+2}$$

$$= \frac{\sin \left((n+1) \frac{2\pi}{n+2} \right)}{\sin \left(\frac{\pi}{n+2} \right)} \cdot \cos \left(\frac{\frac{3\pi}{n+2} + (2n+3) \frac{\pi}{n+2}}{2} \right)$$

$$= \frac{\sin \left((n+1) \frac{\pi}{n+2} \right)}{\sin \left(\frac{\pi}{n+2} \right)} \cdot \cos \left((n+3) \frac{\pi}{n+2} \right) [\sin(\pi - \theta) = \sin \theta]$$

$$= \cos \left((n+3) \frac{\pi}{n+2} \right) = \cos \left(\pi + \frac{\pi}{n+2} \right) = -\cos \left(\frac{\pi}{n+2} \right)$$

$$\sum_{k=0}^k \cos \left((2k+2) \frac{\pi}{n+2} \right) = \cos \left(\frac{2\pi}{n+2} \right) + \cos \frac{4\pi}{n+2} + \dots + \cos \left((2n+2) \frac{\pi}{n+2} \right)$$

$$\frac{\sin \left((n+1) \frac{2\pi}{n+2} \right)}{\sin \left(\frac{\pi}{n+2} \right)} \cos \left((n+2) \frac{\pi}{n+2} \right)$$

$$= \cos \pi = -1$$

$$f(n) = \cos \frac{(n+1) \cos \left(\frac{\pi}{n+2} \right)}{n+2}$$

$$f(n) = \cos \left(\frac{\pi}{n+2} \right)$$

$$\text{Option : (1) } f(4) = \cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$$

$$(2) f(6) = \cos^{-1} (f(6)) = \frac{\pi}{8}$$

$$\alpha = \tan \frac{\pi}{8}$$

$$\alpha^2 + 2\alpha - 1 = 0$$

$$2\alpha = 1 - \alpha^2$$

$$\frac{2\alpha}{1 - \alpha^2} = 1$$

$$\tan \left(\frac{\pi}{8} \times 2 \right) = 1$$

$$\tan 45^\circ = 1 \text{ true}$$

$$(3) n \rightarrow \infty$$

$$f(n) \rightarrow 1$$

$$(4) f(5) = \cos \left(\frac{\pi}{7} \right)$$

$$\cos^{-1} (f(5)) = \frac{\pi}{7}$$

$$\sin \left(7 \times \frac{\pi}{7} \right) = \sin \pi = 0$$

Q9. The value of $\sec^{-1}\left(\frac{1}{4}\sum_{k=0}^{10}\sec\left(\frac{7\pi}{12}+\frac{k\pi}{2}\right)\sec\left(\frac{7\pi}{12}+\frac{(k+1)\pi}{2}\right)\right)$ in the interval $\left[-\frac{\pi}{4}, \frac{3\pi}{4}\right]$ equals

Ans. 0

Sol.

$$\begin{aligned} & \frac{1}{4}\sum_{k=10}^{10}\frac{1}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)} \\ &= \frac{1}{4}\sum_{k=10}^{10}\frac{\sin\left(\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right)}{\cos\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\cos\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)} \\ &= \frac{1}{4}\sum_{k=10}^{10}\left(\tan\left((k+1)\frac{\pi}{2}+\frac{7\pi}{12}\right)-\tan\left(\frac{k\pi}{2}+\frac{7\pi}{12}\right)\right) \\ &= \frac{1}{4}\left(\left(\frac{11\pi}{2}+\frac{7\pi}{12}\right)-\tan\frac{7\pi}{12}\right) \\ &= \frac{1}{4}\left(\tan\frac{\pi}{12}+\cot\frac{\pi}{12}\right) \\ &= \frac{1}{4}2\operatorname{cosec}\left(2\times\frac{\pi}{12}\right) \\ &= \frac{1}{4}\times 2\operatorname{cosec}\left(\frac{\pi}{6}\right) \\ &= \frac{1}{4}\times 2\times 2=1 \\ &\sec^{-1}(1)=0 \end{aligned}$$

Q10. Considering only the principal values of the inverse trigonometric functions, the value of $\frac{3}{2}\cos^{-1}\sqrt{\frac{2}{2+\pi^2}}+\frac{1}{4}\sin^{-1}\frac{2\sqrt{2}\pi}{2+\pi^2}+\tan^{-1}\frac{\sqrt{2}}{\pi}$ ____ is.

Ans. 2.36

Sol.

$$\begin{aligned} \cos^{-1}\sqrt{\frac{2}{2+\pi^2}} &= \tan^{-1}\frac{\pi}{\sqrt{2}} \\ \sin^{-1}\left(\frac{2\sqrt{2}\pi}{2+\pi^2}\right) &= \sin^{-1}\left(\frac{2\times\frac{\pi}{\sqrt{2}}}{1+\left(\frac{\pi}{\sqrt{2}}\right)^2}\right) \\ &= \pi-2\tan^{-1}\left(\frac{\pi}{\sqrt{2}}\right) \end{aligned}$$

$$\begin{aligned}
 & \left(\text{As, } \sin^{-1} \left(\frac{2x}{1+x^2} \right) = \pi - 2\tan^{-1}x, x \geq 1 \right) \\
 & \text{and } \tan^{-1} \frac{\sqrt{2}}{\pi} = \cot^{-1} \left(\frac{\pi}{\sqrt{2}} \right) \\
 & \therefore \text{Expression } \frac{3}{2} \left(\tan^{-1} \frac{\pi}{\sqrt{2}} \right) + \frac{1}{4} \left(\pi - 2\tan^{-1} \frac{\pi}{\sqrt{2}} \right) + \cot^{-1} \left(\frac{\pi}{\sqrt{2}} \right) \\
 & = \left(\frac{3}{2} - \frac{2}{4} \right) \tan^{-1} \frac{\pi}{\sqrt{2}} + \frac{\pi}{4} + \cot^{-1} \frac{\pi}{\sqrt{2}} \\
 & = \left(\tan^{-1} \frac{\pi}{\sqrt{2}} + \cot^{-1} \frac{\pi}{\sqrt{2}} \right) + \frac{\pi}{4} \\
 & = \frac{\pi}{2} + \frac{\pi}{4} = \frac{3\pi}{4} \\
 & = 2.35 \text{ or } 2.36
 \end{aligned}$$

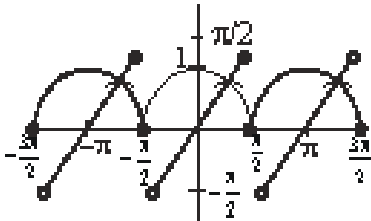
Q11. Let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, for $x \in \mathbb{R}$. Then the number of real solutions of the equation $\sqrt{1 + \cos(2x)} = \sqrt{2}\tan^{-1}(\tan x)$ in the set $\left(-\frac{3\pi}{2}, -\frac{\pi}{2}\right) \cup \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ is equal to _____.

Ans. 3

Sol. $\sqrt{1 + \cos 2x} = \sqrt{2}\tan^{-1}(\tan x)$

$$\sqrt{2\cos^2 x} = \sqrt{2}\tan^{-1}(\tan x)$$

$$\sqrt{2}|\cos x| = \sqrt{2}\tan^{-1}(\tan x)$$



$$y = |\cos x| \quad y = \tan^{-1}(\tan x)$$

3 Solution

Q12. For any $y \in \mathbb{R}$ let $\cot^{-1}(y) \in (0, \pi)$ and $\tan^{-1}(y) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the sum of all the solutions of the equation $\tan^{-1} \left(\frac{6y}{9-y^2} \right) + \cot^{-1} \left(\frac{9-y^2}{6y} \right) = \frac{2\pi}{3}$ for $0 < |y| < 3$, is equal to :

(A) $2\sqrt{3} - 3$ (B) $3 - 2\sqrt{3}$ (C) $4\sqrt{3} - 6$ (D) $6 - 4\sqrt{3}$

Ans. C

Sol. Case-I : $y \in (-3, 0)$

$$\tan^{-1} \left(\frac{6y}{9-y^2} \right) + \pi + \tan^{-1} \left(\frac{6y}{9-y^2} \right) = \frac{2\pi}{3}$$

$$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = -\frac{\pi}{3}$$

$$y^2 - 6\sqrt{3}y - 9 = 0 \Rightarrow y = 3\sqrt{3} - 6 (y \in (-3, 0))$$

Case-I : $y \in (0, 3)$

$$2\tan^{-1}\left(\frac{6y}{9-y^2}\right) = \frac{2\pi}{3} \Rightarrow \sqrt{3}y^2 + 6y - 9\sqrt{3} = 0$$

$$y = \sqrt{3} \text{ or } y = -3\sqrt{3} \text{ (rejected)}$$

$$\text{sum} = \sqrt{3} + 3\sqrt{3} - 6 = 4\sqrt{3} - 6$$

Exercise C1

Q1. The principal value of $\sin^{-1} \left[3 \left[\sin^4 \left(\frac{3\pi}{2} - \alpha \right) + \sin^4(3\pi + \alpha) \right] - 2 \left[\sin^6 \left(\frac{\pi}{2} + \alpha \right) + \sin^6(5\pi + \alpha) \right] \right]$ is equal to
(A) $-\frac{2\pi}{3}$ (B) $\frac{\pi}{2}$ (C) $\frac{4\pi}{3}$ (D) $\frac{5\pi}{3}$

Ans. B

Sol. -

Q2. If $\sin(\cot^{-1}(x+1)) = \cos(\tan^{-1} x)$, then $x =$

(A) $-\frac{1}{2}$ (B) $\frac{1}{2}$ (C) 0 (D) $\frac{9}{4}$

Ans. A

Sol. -

Q3. Sum of series $\cot^{-1} \left(\frac{5}{\sqrt{3}} \right) + \cot^{-1} \left(\frac{9}{\sqrt{3}} \right) + \cot^{-1} \left(\frac{23}{\sqrt{3}} \right) + \dots$ upto infinite terms is equal to

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{6}$ (D) $\frac{\pi}{12}$

Ans. B

Sol. -

Q4. If $2 \tan^{-1} \left(\frac{1+x}{1-x} \right) = 5 \tan^{-1} x$, then value of x is -

(A) $\tan \left(\frac{1}{\sqrt{3}} \right)$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{\pi}{4}$ (D) $\sqrt{3}$

Ans. B

Sol. Put $x = \tan \theta \Rightarrow -\frac{\pi}{2} < \theta < \frac{\pi}{2}$

$$2 \tan^{-1} \left(\frac{1 + \tan \theta}{1 - \tan \theta} \right) = 5 \tan^{-1}(\tan \theta)$$

$$2 \tan^{-1} \left(\tan \left(\frac{\pi}{4} + \theta \right) \right) = 5\theta$$

$$-\frac{\pi}{4} < \frac{\pi}{4} + \theta < \frac{3\pi}{4}$$

Case : I

Case : II

$$\begin{aligned}
-\frac{\pi}{4} &< \frac{\pi}{4} + \theta < \frac{\pi}{2} & \frac{\pi}{2} &< \frac{\pi}{4} + \theta < \frac{3\pi}{4} \\
\Rightarrow -\frac{\pi}{2} &< \theta < \frac{\pi}{4} & x &< 1 \\
x &< 1 & 2\left(\frac{\pi}{4} + \theta - \pi\right) &= 5\theta \\
\frac{\pi}{2} + 2\theta &= 5\theta & 3\theta &= -\frac{3\pi}{2} \\
\theta &= \frac{\pi}{6} & \theta &= -\frac{\pi}{2} \\
x &= \frac{1}{\sqrt{3}} & \Rightarrow & \text{Not defined} \\
\Rightarrow x &= \frac{1}{\sqrt{3}}
\end{aligned}$$

Q5. If $x \in (0,1)$, then solution of inequation $\sin(2 \sin^{-1} x) > \sqrt{2}x$ is given by-

(A) $x \in \left(\frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}\right)$ (B) $x \in \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ (C) $(0, 1)$ (D) $\left(0, \frac{1}{\sqrt{2}}\right)$

Ans. D

Sol. -

Q6. If $\sin^{-1}(x^2) = \cos^{-1} x$, then $x^4 + x^2$ is

(A) $\frac{1}{4}$ (B) 1 (C) $\frac{1}{2}$ (D) $\frac{2}{5}$

Ans. B

Sol. Put $\cos^{-1} x = \sin^{-1} \sqrt{1-x^2}$
then $x^2 = \sqrt{1-x^2}$
 $x^4 + x^2 = 1$

Q7. Which of the following functions represent identical graphs in x-y plane $\forall x \in (2, 3)$?

(A) $y = \cos^{-1} \sqrt{3-x}$ (B) $y = \sin^{-1} \sqrt{x-2}$ (C) $y = \cot^{-1} \left(\sqrt{\frac{3-x}{x-2}}\right)$
(D) $y = \frac{1}{2} \sin^{-1} \left(2\sqrt{(3-x)(x-2)}\right)$

Ans. A, B, C

Sol. -

Q8. Let x_1, x_2, x_3, x_4 be four non zero numbers satisfying the equation

$$\tan^{-1} \frac{a}{x} + \tan^{-1} \frac{b}{x} + \tan^{-1} \frac{c}{x} + \tan^{-1} = \frac{\pi}{2}$$

then which of the following relation(s) hold good?

(A) $\sum_{i=1}^4 x_i = a + b + c + d$ (B) $\sum_{i=1}^4 \frac{1}{x_i} = 0$ (C) $\prod_{i=1}^4 x_i = abcd$

(D) $(x_1 + x_2 + x_3)(x_2 + x_3 + x_4)(x_3 + x_4 + x_1)(x_4 + x_1 + x_2) = abcd$

Ans. D, C, B

Sol. -

Q9. If $x^2 + 2x + n > 10 + \sin^{-1}(\sin 9) + \tan^{-1}(\tan 9)$ for all real x , then the possible value of n can be

(A) 11 (B) 12 (C) 13 (D) 14

Ans. B, C, D

Sol. -

Q10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos^{-1}(-\{x\})$ where $\{x\}$ is fractional part function. Then which of the following is/are correct?

(A) f is many one but not even function.

(B) Range of f contains two prime numbers. (C) f is aperiodic.

(D) Graph of f does not lie below x -axis.

Ans. D, B, A

Sol. -

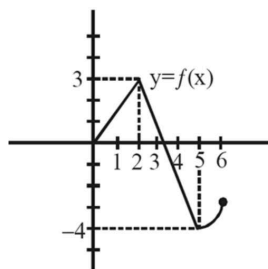
Q11. The value of $\sum_{p=1}^{20} \sum_{q=1}^{20} \tan^{-1}\left(\frac{q}{p}\right)$ is

(A) 200π (B) 220π (C) $\sum_{p=1}^{20} \sum_{q=1}^{20} \left(\tan^{-1} \frac{p}{q} + \tan^{-1} \frac{q}{p} \right)$ (D) 400π

Ans. C, A

Sol. -

Q12. The graph shows a part of an even function $f: [-6, 6] \rightarrow \mathbb{R}$, then-



(A) Domain of $f(|x| + 3)$ is $[-3, 3]$ (B) Range of $f(|x| + 3)$ is $\left[-4, \frac{2}{3}\right]$

$$(C) \cos^{-1} \left(f \left(\frac{1}{3} \right) \right) + \cos^{-1} \left(f \left(\frac{43}{14} \right) \right) = \frac{2\pi}{3}$$

(D) Number of solutions of the equation $f(f(x)) = 2$ is 4.

Ans. A, B, C

Sol. -

Q13. Which of the following statement is/are true

$$(A) \cos^{-1} (\cos (1 + |\sin x|)) = 1 + |\sin x| \text{ for all real } x$$

$$(B) \sin^{-1} (\sin ((\cos^2 x - \sin^2 x))) = \cos^2 x - \sin^2 x \forall x \in \mathbb{R}.$$

$$(C) \tan^{-1} (\tan (x^2 + 2x + 2)) = x^2 + 2x + 2 \forall x \in \mathbb{R}$$

$$(D) \tan^{-1} (\tan (3 \sin x)) = 3 \sin x \forall x \in \mathbb{R}$$

Ans. A, B

Sol. -